

SAUDI ARABIA EMPLOYEE ATTRITION

4/13/2026

DATASET

1.PROJECT OVERVIEW

Employee attrition is a significant challenge for organizations as it impacts productivity, employee morale, and overall business performance. Understanding the factors that influence employee turnover helps organizations develop effective retention strategies and improve workforce stability.

This project focuses on analyzing employee attrition using an HR dataset containing employee demographic information, job and compensation details, training records, and workplace satisfaction factors. The dataset was cleaned and transformed using Microsoft Excel and Power Query by removing duplicates, handling missing values, standardizing categorical data, and creating calculated columns such as Age Category, Salary Level, Attrition Flag, and Useful Training Flag.

To improve data organization and analysis, the cleaned dataset was divided into four logical tables: **Employee Demographics**, **Job and Compensation**, **Training Details**, and **Satisfaction and Attrition**. These tables were connected using **One-to-One (1:1)** relationships based on the **Employee_ID** field in Power BI.

A single interactive dashboard was developed in Microsoft Power BI to provide a comprehensive view of employee attrition. The dashboard enables HR professionals to monitor workforce trends, identify the factors influencing employee turnover, and support data-driven decision-making through interactive visualizations and dynamic filters.

2. OBJECTIVES

The project was designed to achieve the following objectives:

1. **Analyze the overall employee attrition status** by examining workforce size, attrition count, attrition rate, and average salary.
2. **Identify departments and job roles with the highest employee attrition** to determine areas requiring focused retention strategies.
3. **Evaluate the impact of salary, work experience, and overtime on employee attrition** to understand how compensation and workload influence employee turnover.
4. **Analyze the effectiveness of employee training programs** in improving employee retention and reducing attrition.
5. **Assess employee satisfaction and workplace well-being** by examining job satisfaction, physical stress, and their relationship with employee attrition.
6. **Identify the key organizational factors influencing employee attrition** by analyzing promotion opportunities, rewards and wages satisfaction, job support, job engagement, and other workplace factors.
7. **Enable interactive employee attrition analysis** through dynamic dashboard filters, allowing users to explore workforce trends across departments, job roles, gender, age groups, salary levels, and experience categories.

3. DATA SOURCE

Source Description and Timeline:

<https://data.mendeley.com/datasets/6z2hty8php/1/files/b422d0af-74b6-47b8-94ff-614e09c2382f> and 19.03.2025.

Domain: Human Resource Analytics

4.PROBLEM STATEMENT

Employee attrition is a major challenge for organizations, leading to increased recruitment costs, loss of skilled talent, and reduced productivity. This project aims to analyze employee data from various sectors in Saudi Arabia to identify the key factors influencing employee attrition.

There is no clear visibility into which factors contribute most to employees leaving the organization, such as job satisfaction, workload and job roles. The lack of interactive analysis makes it difficult to compare attrition across departments, roles, and demographics.

These factors are not directly structured for analysis and require cleaning, transformation, and feature engineering. The problem is to determine how variables such as salary, job role, department, training, stress levels, and job satisfaction contribute to employee attrition and to provide actionable insights for improving employee retention strategies.

5. ATTRIBUTE DETAILS

Attribute Name	Data Type	Description
Employee_ID	Integer	Unique ID assigned to each employee
Attrition	Text (Yes/No)	Indicates whether employee left the company
Gender	Text	Gender of employee
Age	Text (Categorical)	Age group of employee
Marital_Status	Text	Marital status (Single/Married)
Academic_Degree	Text	Highest education qualification
Years_Experience	Text	Total years of work experience
Sector	Text	Industry sector of employee
Department	Text	Department where employee works
Job_Title	Text	Employee job role
Monthly_Salary	Text	Salary range of employee
Allowances	Integer	Count of Additional allowance amount
Bonus	Text (Yes/No)	Whether employee receives bonus
Over Time	Text (Yes/No)	Overtime work status
Rewards_Wages_Satisfaction	Text (Yes/No)	Satisfaction with salary and rewards
Get_Deserved_Promotion	Text (Yes/No)	Whether employee feels promotion is deserved

Training_Programs	Text	Number/range of training programs attended
Useful_Training_Programs	Text (Yes/No)	Whether training was useful
Job_Support	Text	Level of support at workplace
Emotional_Commitment	Text	Emotional attachment to job
Job_Engagement	Text	Level of engagement at work
Distance_To_Work	Text	Distance from home to workplace
Work_Life_Balance	Text	Work-life balance level
Physical_Stress	Text	Level of physical stress

6.TOOLS & TECHNOLOGIES

Microsoft Excel:

- Data Cleaning
- Performed logical functions and Statistical functions
- Applied conditional formatting

Power Query:

- Removing Duplicates
- Replaced null values
- Datatype Formatting
- Created calculated columns like Age category, Salary Numeric
- Renamed and Merged the columns

Power BI:

- Data Modeling
- DAX calculations
- Visualizations
- Interactive dashboard creation

7. DATA PRE-PROCESSING (EXCEL / POWER QUERY)

Tasks Performed in Excel: Logical functions like IF () and Nested IF () is used to calculate the numerical columns for

- Age - Age_mid
- Attrition – Attrition_flag
- Marital_Status - Marital_status_code
- Distance_To_Work – distance_code
- Seniority

- Salary_numeric - Salary_level
- Exp_numeric - seniority

Conditional Formatting applied for the Attrition column

Statistical functions were used to calculate total count, Attrition count, Attrition rate, Attrition based on many categories.

Tasks Performed in Power Query:

- Duplicates on Employee_ Id (Primary key) are removed
- Replaced the inconsistent values and Standardized the text format
- Merged Sector and Department column into one (Sector-Department)
- Datatype formatting done for each column
- Created calculated columns for Salary, Experience and Age Category since they were categorical

8. DATA MODELLING AND DAX (POWER BI)

Data Modeling:

To improve data organization and simplify analysis, the cleaned employee attrition dataset was divided into four logical tables:

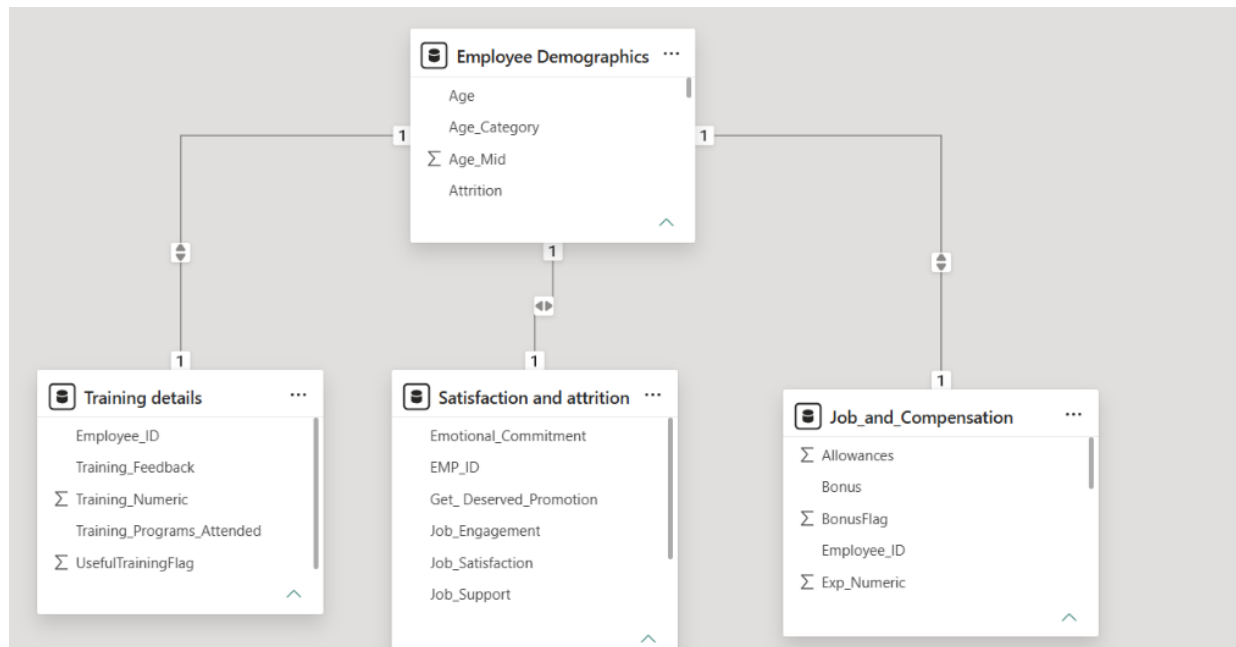
- **Employee Demographics**
- **Job & Compensation**
- **Training Details**
- **Satisfaction & Attrition**

The **Employee Demographics** table serves as the **central table** and is connected to the other three tables using the **Employee_ID** field. Since each employee has only one record in every table, **One-to-One (1:1)** relationships were established between the Employee Demographics table and each of the related tables. All relationships are active, ensuring accurate filtering and interaction across the dashboard.

This data model enables seamless integration of employee demographics, compensation, training, and satisfaction information, allowing comprehensive employee attrition analysis through a single interactive Power BI dashboard.

Data Preparation

Additional analytical columns such as **Age Category, Attrition Flag, Salary Numeric, Salary Level, Experience Numeric, Seniority, Overtime Flag, Bonus Flag, Training Numeric, and Useful Training Flag** were created **using Microsoft Excel formulas** during the data preparation stage before importing the data into Power BI.



DAX Measures:

Total Employees - total number of employees in the dataset

Attrition Count - total number of employees who have left the organization

Attrition Rate (%) – Percentage of employees who left the organization

Active Employees – Employees currently working in the organization

Avg Salary - Average Salary of all employees

Overtime Employees - number of employees who work overtime

Overtime (%) - percentage of employees who work overtime

Bonus (%) - percentage of employees who received a bonus

Useful Training Count - Counts the number of employees who said training was useful. **Useful**

Training % - percentage of employees who found their training useful out of total employees.

Not Useful Training Count - Counts the number of employees who said training was not useful.

Not Useful Training (%) - percentage of employees who found training not useful out of total employees.

Satisfied Employees - number of employees who are satisfied with their job.

Satisfaction % - Percentage of employees who are satisfied with their job out of total employees.

High Stress Employees - number of employees experiencing high physical stress.

Stress % - Percentage of employees experiencing physical stress out of total employees.

9.ANALYSIS AND VISUALIZATIONS (Power BI)

The Employee Attrition Dashboard was developed in Microsoft Power BI to analyze employee turnover and identify the key factors influencing employee retention. The dashboard integrates employee demographics, job and compensation, training details, and satisfaction information into a single interactive report. The analysis is organized into seven business objectives, each supported by relevant visualizations.

Objective 1: Analyze the Overall Employee Attrition Status

Visualizations

- KPI Cards
 - Total Employees
 - Employees Left
 - Attrition Rate
 - Average Salary

Description

KPI cards provide a quick overview of the organization's workforce by displaying the total number of employees, the number of employees who left, the attrition rate, and the average monthly salary. These metrics help HR managers monitor overall workforce performance and employee turnover.

Objective 2: Identify Departments and Job Roles with the Highest Employee Attrition

Visualizations

- Clustered Bar Chart (Attrition by Job Title)
- Treemap (Employees by Department)

Description

The clustered bar chart compares employee attrition across different job roles, while the treemap displays the distribution of employees across departments. These visualizations help identify departments and job roles with higher employee turnover.

Objective 3: Evaluate the Impact of Salary, Experience, and Overtime on Employee Attrition

Visualizations

- Clustered Column Chart (Salary Level vs Attrition)
- Line Chart (Experience vs Attrition)
- Pie Chart (Overtime Distribution)

Description

These visualizations analyze the relationship between employee compensation, work experience, overtime, and employee attrition. They help determine whether salary levels, years of experience, and overtime contribute to employee turnover.

Objective 4: Analyze the Effectiveness of Employee Training Programs

Visualizations

- Stacked Column Chart (Training Programs vs Attrition)
- Donut Chart (Useful Training Feedback)

Description

These visualizations evaluate employee participation in training programs and whether employees considered the training useful. The analysis helps determine the effectiveness of training programs in improving employee retention.

Objective 5: Assess Employee Satisfaction and Workplace Well-being

Visualizations

- Donut Chart (Job Satisfaction)
- Stacked Bar Chart (Physical Stress vs Attrition)

Description

The dashboard analyzes employee satisfaction and workplace stress levels to understand their relationship with employee attrition. These visualizations help identify areas where organizations can improve employee well-being.

Objective 6: Identify the Key Organizational Factors Influencing Employee Attrition

Visualization

- Key Influencers Visual

Description

The Key Influencers visual automatically identifies the organizational factors that have the greatest impact on employee attrition. Variables such as promotion opportunities, rewards and wages satisfaction, job support, job engagement, overtime, bonus, useful training, and physical stress are analyzed to determine their influence on employee turnover.

Objective 7: Enable Interactive Employee Attrition Analysis Visualization

- Dashboard Slicers

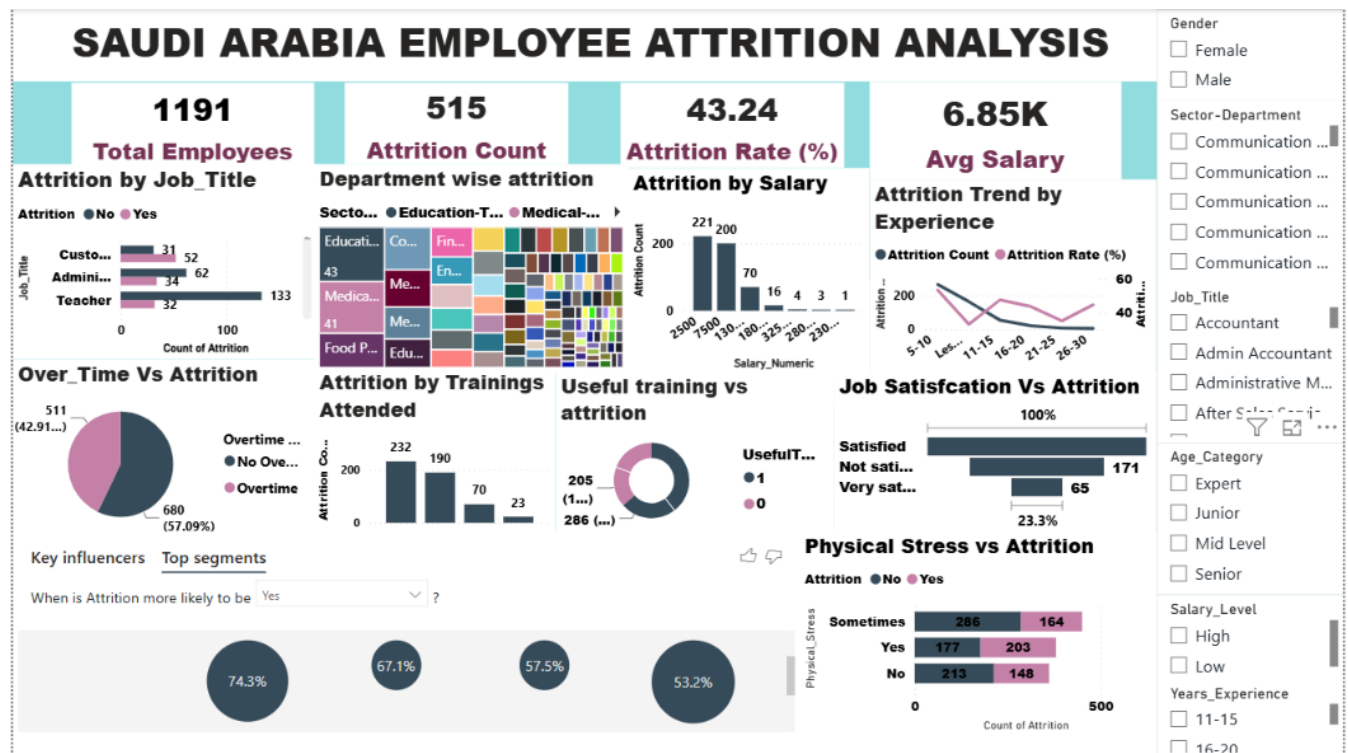
Available Filters

- Gender
- Department
- Job Title
- Age Category
- Salary Level
- Experience

Description

Interactive slicers enable users to dynamically filter the dashboard and analyze employee attrition across different employee groups. This functionality provides flexibility for HR managers to explore workforce trends and make informed decisions.

DASHBOARD SCREENSHOT:



10. KEY FINDINGS & INSIGHTS

Objective 1: Analyze the Overall Employee Attrition Status

KeyInsight:

The dashboard provides a clear overview of the total workforce, employees who left the organization, the overall attrition rate, and the average salary, helping HR managers monitor workforce stability.

Total Employees – 1191, Attrition count - 515, Attrition rate - 43.24%, Avg salary – 6.85K

Objective 2: Identify Departments and Job Roles with the Highest Employee Attrition

KeyInsight:

The analysis highlights the departments and job roles with higher employee attrition, enabling HR teams to identify areas that require focused retention strategies and workforce planning.

Job title - Customer Services Representative =52 (attrition count) May indicate limited career growth

Department – Education- Teaching (43) Shows workload pressure

Objective 3: Evaluate the Impact of Salary, Experience, and Overtime on Employee Attrition

KeyInsight:

The dashboard reveals how salary levels, years of experience, and overtime patterns influence employee attrition, helping organizations understand the impact of compensation and workload on employee retention.

Salary - Less than 5000 – >221, 5000-10000 ->200 (Shows financial dissatisfaction)

Experience – Less than 5 years -> 266, 5-10 years -> 164 (Indicates more job switching opportunities)

Overtime - Yes - 224, No- 291 (Lack of involvement in work)

Objective 4: Analyze the Effectiveness of Employee Training Programs

KeyInsight:

Training participation and employee feedback provide valuable insights into the effectiveness of training programs, supporting improvements in employee development and retention initiatives.

Attended trainings (participation)	1-3 → 232 No Training -> 190	More trainings will reduce the attrition count It improves the engagement
Employee feedback	Useful but not attrited (471)	Not Attrited count is more than attrited one indicates training is effective

Objective 5: Assess Employee Satisfaction and Workplace Well-being

KeyInsight:

The analysis shows the relationship between job satisfaction, workplace stress, and employee attrition, helping organizations identify areas where employee well-being can be improved.

Satisfaction - Satisfied- 279 Satisfied is more than others, shows some other risks
Workplace stress - Yes- 203 High stress indicates more attrition

Objective 6: Identify the Key Organizational Factors Influencing Employee Attrition

KeyInsight:

The Key Influencers analysis identifies organizational factors such as promotion opportunities, rewards, job support, job engagement, and workplace conditions that have the greatest impact on employee attrition.

- Employees **who were not satisfied with rewards and wages were 1.72 times more likely to leave the organization**, making it the strongest factor influencing employee attrition.
- Employees **with difficult job engagement were 1.55 times more likely to experience attrition**, indicating that work engagement plays a significant role in employee retention.
- Employees **who were not satisfied with their jobs were 1.52 times more likely to leave**, highlighting the importance of improving job satisfaction.
- Employees **who did not receive deserved promotions were 1.42 times more likely to leave**, suggesting that career growth opportunities influence employee retention.
- Employees **who did not find training programs useful were 1.40 times more likely to leave**, emphasizing the importance of effective employee development initiatives.
- Employees **experiencing physical stress were 1.39 times more likely to leave**, indicating that workplace well-being has a direct impact on employee attrition.

Objective 7: Enable Interactive Employee Attrition Analysis

KeyInsight:

Interactive dashboard filters allow users to explore employee attrition across different departments, job roles, age groups, salary levels, and experience categories, enabling flexible and data-driven HR decision-making.

Interactive slicers enable users to filter the dashboard by demographic and employment attributes. All KPIs and charts update dynamically, allowing HR managers to analyze employee attrition for specific employee groups and make informed decisions.

11.ANALYTICAL INSIGHTS AND RECOMMENDATIONS

Descriptive Analysis

Employee attrition is higher among employees with low rewards and wages satisfaction, difficult job engagement, low job satisfaction, limited promotion opportunities, ineffective training, and higher physical stress.

Diagnostic Analysis

The primary reasons for employee attrition are dissatisfaction with rewards and wages, poor job engagement, low job satisfaction, lack of deserved promotions, ineffective training programs, and workplace stress.

Predictive Analysis

Employees who are dissatisfied with rewards and wages, have difficult job engagement, receive limited promotions, experience physical stress, or perceive training as ineffective are more likely to leave the organization if these conditions persist.

Prescriptive Analysis

Organizations should improve compensation and rewards, provide fair promotion opportunities, strengthen employee engagement, enhance the effectiveness of training programs, reduce workplace stress, and improve overall job satisfaction to reduce employee attrition and increase employee retention.

10.CONCLUSION:

This project successfully developed an **Employee Attrition Analysis Dashboard** using Microsoft Power BI to understand the factors influencing employee turnover. The dashboard integrates employee demographics, job and compensation, training, and satisfaction data into a single interactive report. The analysis identified key factors such as rewards and wages satisfaction, job engagement, promotions, training effectiveness, and physical stress that influence employee attrition. Interactive visualizations and slicers enable users to explore employee trends and make data-driven decisions. The insights generated can help HR professionals improve employee retention strategies and workplace satisfaction. Overall, this project demonstrates the effectiveness of Power BI in transforming HR data into meaningful and actionable business insights.